

Grade 7 (Form 1) ICT – Exam Revision & Predicted Questions

Based on the 8 units of the MIE Grade 7 ICT workbook: Health, Safety & Ethics · Computer Operation & Fundamentals · Internet · Word Processing · Spreadsheet · Presentation · Multimedia · Practical Problem Solving & Programming.

Note: the book itself could not be opened when this was prepared, so notes follow the standard Grade 7 syllabus for each unit. Check each point against your own copy of the book and your teacher's notes.

Unit 1: Health, Safety and Ethics

- **Ergonomics:** designing the workplace to suit the user. Sit upright, feet flat, screen at eye level about an arm's length away, wrists straight.
- **Health problems:** eye strain, back and neck pain, RSI (repetitive strain injury), headaches. Prevention: take breaks (20-20-20 rule), stretch, use a good chair, adjust screen brightness.
- **Electrical safety:** don't overload sockets, keep liquids away, no loose cables, never touch wires with wet hands, switch off before cleaning.
- **Netiquette:** be polite online, no capital letters (shouting), no cyberbullying, respect others' privacy, think before posting.
- **Cyberbullying:** using ICT to harass or threaten. Keep evidence, don't reply, block, tell a trusted adult/report.
- **Copyright:** legal ownership of creative work. **Plagiarism** = copying work and claiming it as yours. **Piracy** = illegal copying of software. Give credit to sources.
- **Personal data/privacy:** never share password, address, ID; use strong passwords (letters, numbers, symbols, 8+ characters). Don't meet online strangers.
- **Data protection:** Mauritius has a Data Protection Act protecting personal information.

Unit 2: Computer Operation and Fundamentals

- **Computer:** electronic device that accepts input, processes it, stores it and gives output. Cycle: Input → Process → Output (→ Storage).
- **Hardware** = physical parts; **Software** = programs. **System software** (operating system e.g. Windows, Android); **Application software** (Word, Excel, browser).
- **Input devices:** keyboard, mouse, scanner, microphone, webcam, touchscreen, digital camera. **Output devices:** monitor, printer, speakers, projector.
- **Storage:** primary (RAM – temporary/volatile, ROM – permanent); secondary (hard disk, SSD, flash drive, CD/DVD, memory card, cloud). Units: bit < byte < KB < MB < GB < TB (1 byte = 8 bits).
- **CPU** = 'brain' that processes instructions. Other parts: motherboard, RAM, power supply.
- **Types of computers:** desktop, laptop, tablet, smartphone, server, supercomputer.
- **Operating system functions:** manages hardware, files, memory, runs programs, provides user interface (GUI).
- **Files & folders:** create, rename, copy, move, delete, save. File extensions: .docx (Word), .xlsx (Excel), .pptx (PowerPoint), .jpg/.png (image), .mp3 (audio), .mp4 (video).
- **Safe shutdown:** save work, close programs, Shut down from Start menu. **Malware:** virus, worm, spyware – protect with antivirus, updates, backups.

Unit 3: Internet

- **Internet** = worldwide network of connected computers. **WWW** = collection of web pages accessed via the Internet (the web is a service on the Internet).
- **Browser**: Chrome, Firefox, Edge. **Search engine**: Google, Bing. **URL** = web address (e.g. <https://www.example.com>). **ISP** provides Internet access.
- **Search tips**: use keywords, quotation marks for exact phrase, check reliability of source (author, date, .gov/.edu).
- **Email**: parts – To, Cc, Bcc, Subject, Body, Attachment. Advantages: fast, cheap, attachments. Beware of spam and phishing (never click suspicious links).
- **Uses**: communication, research, shopping, banking, entertainment, e-learning. **Social media**: benefits and risks.
- **Downloading** = receiving files from the Internet; **Uploading** = sending files to it. Bookmark = saved web page. Hyperlink = clickable link.
- **Cloud storage**: Google Drive, OneDrive – access files anywhere.
- **Safe use**: strong passwords, don't share personal info, log out on shared computers, look for the padlock (https).

Unit 4: Word Processing

- **Word processor**: software to create, edit, format and print text documents (e.g. MS Word).
- **Formatting text**: font type, size, **bold**, *italic*, underline, colour, alignment (left, centre, right, justify), line spacing.
- **Editing**: cut (Ctrl+X), copy (Ctrl+C), paste (Ctrl+V), undo (Ctrl+Z), redo (Ctrl+Y), save (Ctrl+S), print (Ctrl+P), select all (Ctrl+A), find & replace.
- **Insert**: pictures, shapes, tables, bullets and numbering, page numbers, header and footer, symbols.
- **Page layout**: margins, orientation (portrait/landscape), paper size, columns.
- **Tools**: spell check, thesaurus, word count, print preview.
- **Table**: rows go across, columns go down, cell = intersection of a row and column. Can merge cells, add borders and shading.
- **Advantages over a typewriter**: easy editing, save, spellcheck, formatting, multiple copies.

Unit 5: Spreadsheet

- **Spreadsheet**: software that organises data in rows and columns and does calculations (MS Excel).
- **Terms**: workbook, worksheet, **cell**, **cell reference** (e.g. B3 = column B, row 3), **range** (A1:A5), active cell, formula bar, label, value.
- **Formulas** start with =. Operators: + – * /. Example: =A1+B1.
- **Functions**: =SUM(A1:A5), =AVERAGE(), =MAX(), =MIN(), =COUNT().
- **Formatting**: number format, currency, borders, merge & centre, column width.
- **Charts**: bar/column (compare), line (change over time), pie (parts of a whole). Add title and labels.
- **Sorting** (ascending/descending) and **filtering** data. **Auto-fill** by dragging the fill handle.
- **Advantages**: automatic recalculation when data changes, fewer errors, charts quickly.

Unit 6: Presentation

- **Presentation software** (PowerPoint, Google Slides) creates slides to communicate ideas to an audience.
- **Terms:** slide, layout, theme/design, **transition** (effect moving between slides), **animation** (effect on objects), slide master, slide show, speaker notes.
- **Insert:** text, pictures, shapes, tables, charts, audio, video, hyperlinks.
- **Good design:** few words, bullet points, large readable font (min 24 pt), contrasting colours, 1 idea per slide, consistent theme, relevant images, don't overuse animations.
- **Planning:** purpose, audience, content, storyboard. Start with title slide, end with conclusion/thank-you.
- **Delivering:** practise, face audience, speak clearly, don't just read the slides.
- F5 = start slide show from beginning; Esc = exit show.

Unit 7: Multimedia

- **Multimedia** = combination of text, images, audio, video and animation.
- **Image types:** bitmap/raster (pixels, e.g. JPG, PNG, GIF) and vector (shapes, scalable, e.g. SVG). **Pixel** = smallest picture element. Resolution = number of pixels.
- **Audio formats:** MP3, WAV. **Video formats:** MP4, AVI. **Compression** reduces file size.
- **Editing:** crop, resize, rotate, brightness/contrast, trim audio/video, add transitions and titles.
- **Devices:** digital camera, scanner, microphone, camcorder.
- **Uses:** education, games, advertising, entertainment, virtual reality, websites.
- **Copyright:** use only royalty-free or permitted media and acknowledge the source.

Unit 8: Practical Problem Solving and Programming

- **Problem solving steps:** understand the problem → plan (algorithm) → implement/code → test → fix errors (debug) → evaluate.
- **Algorithm** = step-by-step instructions to solve a problem. Represented by **flowcharts** or **pseudocode**.
- **Flowchart symbols:** oval = Start/End; rectangle = process; parallelogram = input/output; diamond = decision (Yes/No); arrows = flow.
- **Programming concepts:** sequence, selection (if...then...else), repetition (loops), variables (named storage), input/output.
- **Scratch:** block-based visual language. Sprite = character; stage = background; scripts built by snapping blocks; events (when green flag clicked), motion, looks, sound, control, sensing blocks.
- **Bug** = error in a program; **debugging** = finding and fixing bugs. **Syntax error** (wrong rules) vs **logic error** (wrong result).
- **Qualities of a good algorithm:** clear, finite, correct, ordered steps.

Key Definitions – Quick Glossary

Term	Meaning
Hardware	Physical parts of a computer
Software	Programs that tell hardware what to do
RAM	Temporary memory; lost when power is off
ROM	Permanent memory; holds start-up instructions
Netiquette	Rules of polite behaviour online
Plagiarism	Copying someone's work and claiming it as your own
Phishing	Fake emails/sites that steal personal data
Malware	Harmful software e.g. virus, spyware
URL	Web address
Cell reference	Column letter + row number, e.g. C4
Transition	Effect when moving from one slide to the next
Pixel	Smallest dot in a digital image
Algorithm	Ordered steps to solve a problem
Debugging	Finding and fixing errors in a program

Predicted Questions

Section A – Multiple Choice (1 mark each)

1. Which is an output device?
A Mouse B Scanner C Printer D Microphone
2. Which unit is the largest?
A KB B MB C GB D Byte
3. Copying someone's work and claiming it as yours is called:
A Piracy B Plagiarism C Hacking D Netiquette
4. Which shortcut copies selected text?
A Ctrl+V B Ctrl+X C Ctrl+C D Ctrl+Z
5. Which cell reference is correct?
A 3B B B3 C B-3 D 3-B
6. A formula in a spreadsheet must begin with:
A + B # C = D @
7. Which chart best shows parts of a whole?
A Line B Pie C Bar D Scatter
8. Which flowchart shape shows a decision?
A Oval B Rectangle C Diamond D Parallelogram
9. The memory that loses its content when power is off:
A ROM B Hard disk C RAM D Flash drive
10. Effects between slides are called:
A Animations B Transitions C Themes D Layouts
11. A program error is called a:
A Bug B Sprite C Loop D Pixel
12. Which file extension is a Word document?
A .xlsx B .pptx C .docx D .mp3
13. Which software is an operating system?
A Excel B Windows C Chrome D Scratch
14. A strong password contains:
A Your name B 1234 C Letters, numbers and symbols D Your birthday
15. Sending a file from your computer to the Internet is:
A Downloading B Uploading C Browsing D Booting

Section B – Short Answer

1. Define hardware and software. Give two examples of each. [4]
2. State **three** health problems caused by long computer use and one way to prevent each. [6]
3. List **four** rules of good netiquette. [4]
4. What is cyberbullying? State two actions a victim should take. [3]

5. Differentiate between RAM and ROM. [4]
6. Name two input devices, two output devices and two storage devices. [6]
7. Differentiate between the Internet and the World Wide Web. [2]
8. Name the parts of an email. State two advantages of email. [5]
9. Explain what phishing is and how you can protect yourself. [3]
10. Explain the terms: cell, row, column, cell reference. [4]
11. Write the spreadsheet formula to add cells B2 to B8, and the formula to find the average of C1 to C10. [2]
12. Give two differences between a line chart and a pie chart. [2]
13. State four rules for designing a good presentation slide. [4]
14. Differentiate between transition and animation. [2]
15. What is multimedia? Name its five components. [3]
16. Differentiate between a bitmap and a vector image. [2]
17. What is an algorithm? Name two ways of representing one. [3]
18. Name the shapes used in a flowchart for: start, process, input/output, decision. [4]
19. What is a bug? Explain debugging. [2]
20. Explain what is meant by copyright and why it matters. [3]

Section C – Practical / Structured Questions

1. **Word processing:** You must type a school notice. List the steps to (a) set the title to 18 pt bold centred, (b) insert a picture, (c) add page numbers in the footer, (d) change orientation to landscape. [8]
2. **Spreadsheet:** A table shows marks of 5 pupils in 3 subjects. (a) Write the formula for each pupil's total. (b) Write the formula for the highest mark in Maths. (c) State which chart you would use to compare pupils and why. (d) Explain what happens to the total if one mark is changed. [8]
3. **Presentation:** Plan a 5-slide presentation on 'Safe Internet Use'. Give the title and contents of each slide and two design tips you would follow. [8]
4. **Flowchart:** Draw a flowchart that asks for a number and prints 'Positive' if it is greater than 0, otherwise 'Not positive'. [6]
5. **Scratch:** Describe the blocks you would use to make a sprite move 10 steps repeatedly until it touches the edge, then say 'Game over'. [5]
6. **Problem solving:** Write an algorithm in steps to make a cup of tea / to find the largest of three numbers. [5]
7. **Scenario:** A classmate posts an embarrassing photo of you online and sends you unkind messages. (a) Identify the problem. (b) State three actions you should take. (c) Which rules of netiquette were broken? [6]

Answer Guide

Section A

1-C, 2-C, 3-B, 4-C, 5-B, 6-C, 7-B, 8-C, 9-C, 10-B, 11-A, 12-C, 13-B, 14-C, 15-B

Section B – Model answers

- 1. Hardware: physical parts (monitor, keyboard). Software: programs (Windows, Word).
- 2. Eye strain – take breaks/adjust brightness; back pain – good posture and chair; RSI – correct typing posture, breaks and stretches.
- 3. Be polite; don't use all capitals; respect privacy; don't spread rumours or bully; think before posting; cite sources.
- 4. Using ICT to harass or threaten others. Save evidence, block the person, report to a trusted adult/website.
- 5. RAM: temporary, read/write, loses data when off. ROM: permanent, stores start-up instructions.
- 7. Internet is the global network; WWW is the collection of web pages accessed through it.
- 8. To, Cc, Bcc, Subject, Body, Attachment. Fast, cheap, can attach files, reach many people.
- 9. Phishing: fake messages that steal data. Don't click unknown links, check sender, never give passwords.
- 10. Cell: box where a row and column meet. Row: horizontal. Column: vertical. Reference: e.g. B3.
- 11. =SUM(B2:B8); =AVERAGE(C1:C10).
- 12. Line: shows change over time; pie: shows share of a whole.
- 13. Few words, large font, contrasting colours, relevant images, consistent theme.
- 14. Transition: between slides. Animation: on objects within a slide.
- 15. Combination of text, images, audio, video and animation.
- 16. Bitmap: made of pixels, loses quality when enlarged. Vector: made of shapes, scales without loss.
- 17. Step-by-step instructions to solve a problem; flowchart and pseudocode.
- 18. Oval; rectangle; parallelogram; diamond.
- 19. Bug: error in a program. Debugging: finding and fixing it.
- 20. Legal ownership of creative work; protects creators; do not copy without permission.

Section C – Hints

- Q4: Start → Input number → Decision 'number > 0?' → Yes: Output 'Positive' / No: Output 'Not positive' → End.
- Q5: When green flag clicked → forever/repeat until (touching edge) { move 10 steps } → say 'Game over'.
- Q2: Total =SUM(B2:D2); Highest =MAX(B2:B6); a column chart compares individuals; totals update automatically.
- Q7: Cyberbullying; keep evidence, block, tell parent/teacher/report; broke respect, privacy and kindness rules.

Exam tips: Read each question twice · Use the marks as a guide to the number of points · Practise shortcuts and formulas · Revise definitions · Draw neat flowcharts · Manage your time.